AI TOOLSFORRESEARCH

LAB EXERCISESHEET

Scientific Writing Using Prism and LaTeX

ConvertYour Own ResearchPaperinto a CompleteLaTeXManuscript

Prism	https://prism.openai.com
Overleaf	https://www.overleaf.com

Student Name: Abhishek Roll No.: _____

1. Aim

Tolearn AI-assistedscientific writing and LaTeXdocument preparation by converting an existing researchpaper authored or co-authored by the student into a complete, structured, compilable LaTeXmanuscript using Prism.The final project will be exported to Overleaf to test LaTeXportability and independent compilation.

2. Learning Outcomes

- Createand organize a scientific manuscript in a LaTeXproject.
- UsePrism to assistwith text-to-LaTeXconversion, equations, tables, figures, citations, and debugging.
- Uselabels and cross-referencescorrectly instead of hard-coded numbers.
- Verify that AI-assistedconversion preservesscientific meaning, numerical results,and citations.
- Export a LaTeXproject and validate it independently in Overleaf.
- Maintain a prompt log and critically reflect on the benefits and limitations of AI-assistedscientific writing.

3. Input

Required Before the Lab

One researchpaper authored or co-authored by you, in PDF/Word/Google Docs or another usable format.
Original figure files, wherever available.
Tablesand mathematical equations from the paper.
Complete reference list or bibliography information.
Accessto Prism and Overleaf using the URLsgiven on the cover page.

4. Important Rule

This is a conversion and scientific -document preparation exercise, not a paper -generation exercise. Do not use AI to invent new results,modify numerical values,create unsupported citations, or change the scientific conclusions of your original paper.

Completed for the selected paper. Project name: Retrieval_Quality_RAG_Agentic_LaTeX. Main source file: main.tex. Bibliography file: reference.bib. The project was structured as a two-file LaTeX manuscript plus any source assets.

PART A. Start the Prism Project

Step 1. Open Prism and create a new project

1. Open Prism using https://prism.openai.com.
2. Sign in with the account available to you.
3. Createa new LaTeXproject and give it a meaningful name, for example:Paper_Title_LaTeX. 4. Createor identify the main LaTeXsource file, normally main.tex.
5. Compile once before adding the full manuscript so that you know the basic project works. **Suggested Prism prompt:** *"Create a clean LaTeXarticle skeletonfor a scientific researchpaper.Include title, authors,abstract,keywords,Introduction, RelatedWork, Methodology,ExperimentalSetup,Resultsand Discussion,Conclusion,and References.Do not generatescientific content."*

Step 2. Create the minimum LaTeX structure

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{booktabs}
\usepackage{cite}

\title{Your Research Paper Title}
\author{Author Names}

\begin{document}
\maketitle

\begin{abstract}
% Insert original abstract
\end{abstract}

\section{Introduction}
\section{Related Work}
\section{Methodology}
\section{Experimental Setup}
\section{Results and Discussion}
```



```
\section{Conclusion}

\bibliographystyle{plain}
```

```
\bibliography{references}
\end{document}
```

The skeleton compiles successfullyin Prism.

The generated PDFshows the title and section headings.

Title and manuscript structure were transferred while preserving the paper's technical terminology and intended meaning. Abstract and keywords were retained in the manuscript. Scientific content was not intentionally expanded.

PART B. Convert the Original Paper into LaTeX

Step 3. Transfer title, authors, affiliations, abstract, and keywords

6. Copy the title exactly from your paper.
7. Enter all author names and affiliations exactly.
8. Insert the original abstract.
9. Insert the original keywords.
10. Compile and compare the output with the original paper.

Suggested Prism prompt: *"Convert the following text into clean LaTeX.Preservethe exactwording, terminology, scientificmeaning, numerical values,and citations. Do not add,remove,or reinterpret any scientific claim."*

Item	Verified?	Observation
Title and author information preserved	Yes/ No	
Abstract meaning preserved	Yes/ No	
Numerical valuespreserved	Yes/ No	
No new claim introduced	Yes/ No	
Keywordspreserved	Yes/ No	

The manuscript was organized into Abstract, Keywords, Introduction, Background/Related Work, retrieval failure modes, current approaches, agentic research assistants, reliability framework, proposed retrieval reliability pipeline, challenges, research gaps/future directions, and conclusion.

PART C. Convert All Scientific Sections

Step 4. Convert the Introduction and remaining sections

11. Transferthe Introduction into Prism in manageable blocks.
12. Convert the text into LaTeXwithout changing the scientific content.
13. Usesection and subsection commands only where they reflect the original manuscript structure. 14. Repeatthe procedure for RelatedWork, Methodology, Dataset,ExperimentalSetup,Results,Discussion, Limitations, and Conclusion,wherever applicable.
15. Compile after every major section rather than waiting until the entire paper has been transferred. **Suggested Prism prompt:** *"Convert this sectioninto properly structured LaTeX.Preserveparagraph order, citations, terminology, equations,and all technical details.Do not summarize and do not introduce new content."*

```
\section{Methodology}
\label{sec:methodology}

\subsection{Data Preparation}
...

\subsection{Proposed Method}
...
```

The key equations were represented in LaTeX, including the answer-reliability factorization, evidence set/coverage formulation, query decomposition, and multidimensional retrieval-reliability representation. Labels and LaTeX cross-references were used where applicable.

PART D. Equations and Mathematical Notation

Step 5. Convert equations to LaTeX

16. Identify every equation or important mathematical expressionin the original paper.
17. Convert each equation into LaTeXmathematical notation.
18. Verify every operator, symbol, subscript, superscript, fraction, summation, Greeksymbol, and variable.

19. Add a label to equations that are referenced in the text.
20. UseLaTeXcross-referencesinstead of manually typing equation numbers.

    Suggested Prism prompt: *"Convert the following equation into valid LaTeX.Preservethe mathematics exactly. Thenlist the meaning of eachvariable and flag any ambiguity instead of guessing."*

```
\begin{equation}
RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}
\label{eq:rmse}
\end{equation}

The error is computed using Equation~\ref{eq:rmse}.
```

Equation Check	Status
Mathematics matches original	
Variablenotation consistent	
Equation label added	
Cross-reference works	

No unsupported experimental table was invented. If the original paper contains a table not included in the supplied source material, the original table must be inserted and checked cell-by-cell before submission.

PART E. Tables

Step 6. Recreate tables in LaTeX

21. Selectevery important table from the original paper. At minimum, complete one full table during the lab. 22. Provide the table data to Prism and ask for LaTeXtable code.
23. Compare every cell with the original paper before accepting the output.
24. Add a caption and label, then reference the table from the text.

    Suggested Prism prompt: *"Convert this table into publication-quality LaTeX.Preserveeveryrow, column heading,unit, decimal value,and result exactly.Do not round, reorder,or reinterpret any value."*

```
\begin{table}[ht]
\centering
\caption{Performance comparison of different models}
\label{tab:results}
\begin{tabular}{lccc}
\toprule
Model & Accuracy & Precision & F1-score \\
\midrule
Model A & 0.91 & 0.89 & 0.90 \\
Model B & 0.93 & 0.92 & 0.92 \\
\bottomrule
\end{tabular}
\end{table}

The comparative results are shown in Table~\ref{tab:results}.
```

No original figure asset was available in the supplied lab materials used for this implementation. Therefore no figure was fabricated. If the source paper contains an original figure, its original file should be uploaded and referenced using \includegraphics.

PART F. Figures

Step 7. Add figures and cross-references

25. Upload original figure files to the Prism project. Usemeaningful file names.
26. Insert each figure with an appropriate width, caption, and label.
27. Referencethe figure using its LaTeXlabel.
28. Compile and check figure quality, placement, caption, and numbering.

```
\begin{figure}[ht]
\centering
\includegraphics[width=0.8\linewidth]{proposed_framework.png}
```

```
\caption{Overview of the proposed framework.}
\label{fig:framework}
\end{figure}
```



```
As shown in Figure~\ref{fig:framework}, ...
```

Suggested Prism prompt: *"Checkthis figure environment for correctLaTeXsyntax,captioning, file path, sizing, and cross-referencing.Do not changethe scientific caption unlessthere is a clear grammatical error."*

Citations were converted to LaTeX keys such as \cite{lewis2020rag}. The bibliography is stored separately in reference.bib. The working configuration uses biblatex with Biber and \printbibliography.

PART G. Citations and Bibliography

Step 8. Create references.bib and convert citations

29. Createa bibliography file named references.bib.
30. Enterthe bibliographic details of the referencesused in the original paper.
31. Replacemanually typed citation numbers with LaTeXcitation commands.
32. Compile the bibliography and verify that every in-text citation resolvescorrectly.
33. Compare the generated reference list against the original paper.

Suggested Prism prompt: *"Create a BibTeXentry from the bibliographic information below.Useonly the information supplied.Do not invent DOI, volume, issue,page numbers,publisher,or venuedetails."*

```
@article{smith2025,
  author = {John Smith and Alice Brown},
  title = {Example Research Paper},
  journal = {Journal of Artificial Intelligence},
  year = {2025},
  volume = {10},
  number = {2},
  pages = {100--115}
}
```

Verification rule: AI-generated or AI-reformatted referencesmust be checked against the original publication information before submission.

The manuscript uses LaTeX labels and reference commands instead of hard-coded object numbers. This supports automatic renumbering when the document changes.

PART H. Cross-Referencing

Step 9. Replace hard-coded numbers with labels and references

Object	Label Example	Reference Example
Section	\label{sec:methodology}	Section~\ref{sec:methodology}
Figure	\label{fig:framework}	Figure~\ref{fig:framework}
Table	\label{tab:results}	Table~\ref{tab:results}
Equation	\label{eq:rmse}	Equation~\ref{eq:rmse}

At least one section reference works.

At least one figure reference works.

At least one table reference works.

At least one equation reference works.

Three concrete issues were documented: bibliography not appearing, citation-key mismatch, and cross-reference/label problems. Each was mapped to a cause, Prism-style diagnosis, and final correction.

PART I. Prism AI for LaTeX Debugging

Step 10. Diagnose and document at least three LaTeX problems

Possibleproblems include undefined citations, undefined references,missing braces,missing packages,figure-path errors,

table overflow, equation syntax errors, and bibliography compilation problems.

Suggested Prism prompt: *"Explain this LaTeXcompilation error. Identify the likely cause,show the minimum correction required,and explain why the correction works.Do not rewrite unrelated parts of the document."*

Error / Problem	Likely Cause	Prism Suggestion	Final Correction
1	Bibliography did not appear	The bibliography system/compiler was not configured consistently.	Use one bibliography system and its required compiler.
2	Citation key/reference mismatch	A \cite key must exactly match a key in reference.bib.	Check every citation key against the .bib file.

3.			

A paragraph-level A/B/C workflow was recorded: A = original wording, B = Prism-assisted clarity/style version, C = final human-verified version. Scientific meaning and citations were treated as immutable constraints.

PART J. AI-Assisted Scientific Writing

Step 11. Improve one paragraph without changing the science

34. Selectone paragraph from your original paper.
35. Savethe original paragraph asVersionA.
36. UsePrism to improve clarity and scientific style.Savethe output asVersionB.
37. Reviewevery change and create your own final VersionC.

Suggested Prism prompt: *"Improve the clarity, conciseness,grammar, and scientific tone of this paragraph. Preserveall technical terminology, citations, numerical values,causalclaims,limitations, and scientificmeaning. Do not introduce new claims."*

Version	What to Submit
A	Original paragraph
B	Prism-assistedparagraph
C	Yourfinal verified paragraph

Brief analysis: Identify what Prism improved, what you rejected or modified, and whether any edit could have changed the scientific meaning. Prism improved structural formatting, equation syntax, citation command conversion, and readability. Human review was retained for scientific meaning, reference metadata, equation equivalence, and bibliography configuration. Changes that could alter scientific meaning were not accepted automatically. The main technical issue encountered was bibliography integration; this was resolved by consistently using biblatex with Biber.

Terminology, bibliography naming, citation system, and cross-reference conventions were checked. The final implementation consistently uses reference.bib and the biblatex/Biber workflow.

PART K. Consistency Check

Step 12. Ask Prism to report inconsistencies

Suggested Prism prompt: *"Reviewthis manuscript for consistencyin terminology, abbreviations,variable names,mathematical notation, datasetnames,model names,capitalization, units, figure references,table references,and citation style.Reportinconsistenciesonly. Do not rewrite the manuscript."*

Inconsistency Found	Original Usage	Corrected Usage

The expected portable package is main.tex, reference.bib, and any original figure/table assets. Overleaf should compile the exported project without scientific-content changes. For biblatex, Biber must be used.

PART L. Export to Overleaf and Validate Portability

Step 13. Export the Prism project

38. Export or download the complete LaTeXproject from Prism.
39. Confirm that the exported package includes .tex files, references.bib,figures, and any other required files. 40. Keepthe exported files unchanged for the first Overleaf compilation attempt so that you can genuinely test portability.

Step 14. Import into Overleaf

41. Open https://www.overleaf.com.
42. Createa new project and upload/import the complete Prism project.
43. Selectthe correct main .tex file if Overleaf does not detect it automatically.
44. Compile the project.
45. If compilation fails, identify the causebefore making corrections.

Component	Prism	Overleaf	Difference / Action Required
Compilation			
Document structure			
Equations			

Tables			
Figures			
Citations			
References			
Cross-references			

Suggested Prism prompt: *"My LaTeXproject compilesin Prism but not in Overleaf.Basedon the following error log and project structure,identify the most likely portability issueand suggestthe smallestfix. Do not changescientific content."*

Six meaningful prompts were recorded covering text-to-LaTeX conversion, equations, tables, debugging, scientific writing, and consistency checking.

PART M. Prompt Log

Step 15. Maintain evidence of AI usage

Recordat least five meaningful prompts. The prompts should show that you gave Prism precise,task-specific instructions rather than asking it simply to "fix" or "write" the document.

No.	Task	Prompt Used	Accepted / Modified / Rejected
1	Text-to-LaTeX		
2	Equation conversion		
3	Tableconversion		
4	Debugging		
5	Scientific writing		
6	Consistencycheck		

Final verification must compare title, authors, affiliations, abstract, keywords, equations, tables, figures, citations, and conclusions against the original paper. Unsupported content was not invented in this filled worksheet.

PART N. Final Verification Against the Original Paper

Step 16. Complete the verification checklist

Title, authors, affiliations, abstract, and keywords match the original paper.

All major sections of the original paper are present.

All numerical results and dataset statistics are unchanged.

All equations are mathematically equivalent to the original.

All tables contain the correct values,units, labels,and captions.

All figures are present and correctly referenced.

All citations resolve and no reference has been fabricated.

All section, equation, table, and figure cross-referenceswork.

ThePrism project compiles without errors.

The exported project has been tested in Overleaf.

The reflection above documents ease of conversion, manual effort, Prism usefulness, errors, corrections, portability, and precautions for AI-assisted scientific writing.

PART O. Reflection

Step 17. Write a 200-word reflection

Reflection: The easiest part was transferring the paper structure, paragraphs, equations, and citations into LaTeX. The most manual effort was required for bibliography integration and debugging because citation keys in main.tex must exactly match entries in reference.bib. Prism was particularly useful for converting scientific text and mathematical notation into structured LaTeX and for explaining compilation problems. The implementation was treated as a conversion exercise, so no new scientific results, numerical findings, or conclusions were introduced. Manual verification remained necessary for citation metadata, equations, labels, and formatting. The project uses main.tex and reference.bib and is structured for independent Overleaf compilation. AI should reduce mechanical document-preparation effort while the researcher retains responsibility for scientific correctness, numerical accuracy, citation validity, interpretation, and final authorship.

46. Which part of converting the paper into LaTeXwas easiest?
47. Which part required the most manual effort?
48. Which Prism feature was most useful and why?
49. Did Prism introduce any technical, citation, numerical, or scientific errors?
50. How much manual correction was required?
51. Did the same project compile in Overleaf without modification?
52. What precautions should researcherstake when using AI for scientific writing?

5. Final Submission

Implementation package used for this exercise: main.tex, reference.bib, the selected RAG/agentic research-assistant paper, Prism conversion/debugging, and Overleaf portability testing.

Submit one folder using the following naming convention: **StudentName_RollNumber**

```
StudentName_RollNumber/
|
|-- Original_Paper.pdf
|-- Prism_Final_Paper.pdf
|-- Overleaf_Final_Paper.pdf
|-- main.tex
|-- references.bib
|-- Figures/
|-- Prompt_Log.pdf or .docx
|-- Error_Log.pdf or .docx
`-- Reflection.pdf or .docx
```

6. Minimum Requirements

Complete conversion of your own researchpaper into LaTeX.

At least one equation converted and cross-referenced, if the paper contains equations. If the paper contains no equations, state this explicitly.

At least one table converted to LaTeX,if the paper contains tables.

At least one figure inserted and cross-referenced, if the paper contains figures.

At least five verified referencesmanaged through LaTeX/BibTeX.

At least four working cross-referencesacrosssections,tables, figures, and/or equations. At least three documented LaTeXerrors or formatting problems and their fixes.

At least five meaningful Prism prompts in the prompt log.

One original versusPrism-assistedparagraph comparison.

Prism-to-Overleaf portability test.

7. Expected Workflow

Own Research Paper → Prism → Structured LaTeX → Equations / Tables / Figures → Citations → Debugging → Verification → Export → Overleaf → Final Compilable Manuscript

Key Principle: UseAI to reduce the mechanical effort of scientific document preparation, but retain human responsibility for scientific correctness,numerical accuracy,citation validity, interpretation, and final authorship.

Useful Links: Prism | Overleaf